

Adversarial Paraphrasing: A Universal Attack for Humanizing AI-Generated Text

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🔗 **Project:** <https://github.com/chengez/Adversarial-Paraphrasing>

Abstract

The increasing capabilities of Large Language Models (LLMs) have raised concerns about their misuse in AI-generated plagiarism and social engineering. While various AI-generated text detectors have been proposed to mitigate these risks, many remain vulnerable to simple evasion techniques such as paraphrasing. However, recent detectors have shown greater robustness against such basic attacks. In this work, we introduce **Adversarial Paraphrasing**, a training-free attack framework that universally humanizes any AI-generated text to evade detection more effectively. Our approach leverages an off-the-shelf instruction-following LLM to paraphrase AI-generated content under the guidance of an AI text detector, producing adversarial examples that are specifically optimized to bypass detection. Extensive experiments show that our attack is both broadly effective and highly transferable across several detection systems. For instance, compared to simple paraphrasing attack—which, ironically, increases the true positive at 1% false positive (T@1%F) by 8.57% on RADAR and 15.03% on Fast-DetectGPT—adversarial paraphrasing, guided by OpenAI-RoBERTa-Large, reduces T@1%F by 64.49% on RADAR and a striking 98.96% on Fast-DetectGPT. Across a diverse set of detectors—including neural network-based, watermark-based, and zero-shot approaches—our attack achieves an average T@1%F reduction of 87.88% under the guidance of OpenAI-RoBERTa-Large. We also analyze the tradeoff between text quality and attack success to find that our method can significantly reduce detection rates, with mostly a slight degradation in text quality. Our adversarial setup highlights the need for more robust and resilient detection strategies in the light of increasingly sophisticated evasion techniques.

1 Introduction

Recent advancements in natural language generation have given rise to transformer-based Large Language Models (LLMs) such as GPT [23], Gemini [7], and LLaMA [21], which have demonstrated remarkable capabilities across a wide range of tasks, such as email composition and code generation. These models are capable of producing fluent, coherent text that can be difficult to distinguish from that written by humans. However, despite their impressive performance, LLMs also raise significant security and ethical concerns, including risks related to plagiarism and social engineering.

To counter these risks, the development of reliable AI-generated text detection tools has become a critical research problem. Several works have proposed training neural network-based classifiers to address this challenge [10, 8, 26, 30, 34, 18]. Although typically weaker than trained detectors,

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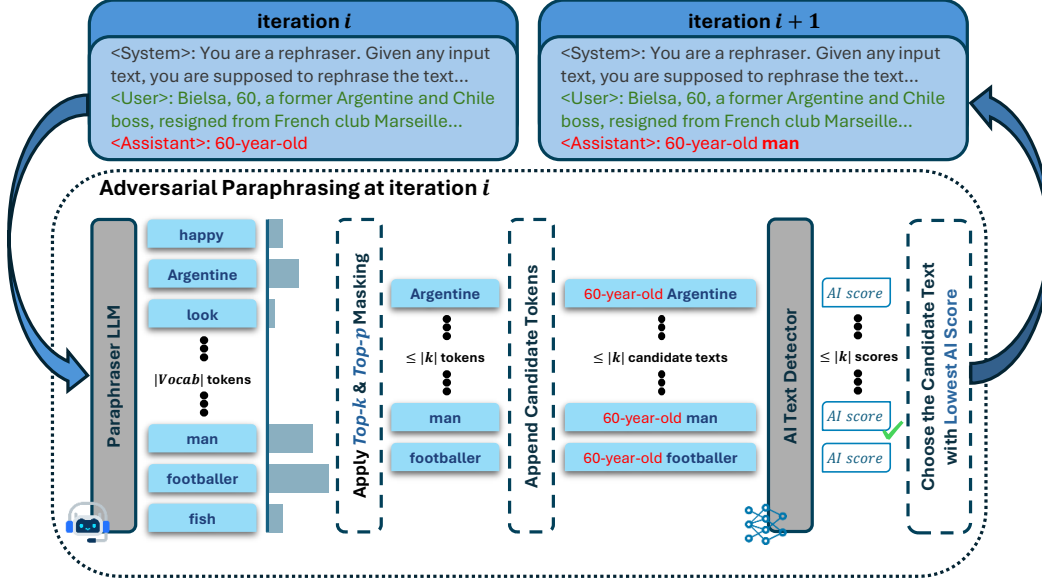


Figure 1: **An overview of our universal and training-free framework for humanizing AI text.** At every auto-regressive step of adversarial paraphrasing, using the guidance from an AI text detector, we search for the token with lowest “AI-score” from the set of top candidate tokens sampled by the paraphraser LLM. The token generation iterations continue until the paraphrasing is finished. (i.e. [EOS] token is sampled)

various zero-shot detection techniques [22, 1, 6, 17] have also been introduced to reduce the overhead of training classifiers. Another propitious approach is watermarking LLMs, which enforces specific signatures in the output text to facilitate detection [2, 14, 16, 36, 4]. While these detection methods show promise in specific contexts, concerns about their robustness remain.

Sadasivan *et al.* [28] and Krishna *et al.* [15] demonstrated how AI-generated text can be paraphrased by another AI model to evade detection by the existing AI detectors. The former also showed that recursively paraphrasing AI-generated content can be used to stress-test these systems and highlighted the fundamental difficulty of reliably detecting AI-generated text. These paraphrasing-based attacks aim to obscure the statistical patterns and artifacts that detection models typically rely on. However, more recent work suggests that some advanced detectors — especially those trained on paraphrased AI outputs [8] — may exhibit greater resilience to such evasion techniques. These developments naturally lead to a pressing question: “*Is it possible to develop a universal attack framework that can consistently and effectively bypass these robust AI-generated text detectors with transferability to a wide variety of other detection systems?*”

In this paper, we introduce a universal, training-free framework for humanizing AI-generated text, a novel attack designed to effectively and efficiently rephrase AI-generated text to evade a wide range of detectors with greater success when compared to the existing attack evasion methods. Our approach, **Adversarial Paraphrasing**, leverages an existing instruction fine-tuned LLM (LLaMA-3-8B configured as a paraphraser via a custom system prompt) not just for paraphrasing, but for adversarially humanizing the text under the guidance of a trained AI text detector (see Figure 1). During the paraphrasing process, at each token generation step, we consider the top likely next tokens proposed by the paraphrasing LLM. Then, instead of standard decoding, we score each of these potential continuations using a guidance AI text detector. We then select the token that leads the sequence to be predicted with the lowest “AI score” (or equivalently, the highest “human score”) by the detector, akin to a targeted, detector-guided beam search at depth one. Some example results are shown in Table 1. This method operationalizes the goal of humanization directly within the generation process, falling under the umbrella of controlled text generation [37, 19, 3].

We demonstrate that our adversarial paraphrasing strategy is not only more effective than baseline methods such as simple or recursive paraphrasing, but unlike other methods, it is also universally transferable to other diverse detection methods, including neural network-based, zero-shot, and watermarking-based detectors. This justifies the universality of our framework. For example, while

simple paraphrasing ironically increases true positive rate at 1% false positive rate (T@1%F) by 8.57% on RADAR [8] and 15.03% on Fast-DetectGPT [1], our adversarial approach, guided by OpenAI-RoBERTa-Large [30], reduces T@1%F by 64.49% and a striking 98.96%, respectively. Evaluated across eight detectors from three different categories, our method achieves an average T@1%F reduction of 87.88%, all while maintaining minimal degradation in text quality. We comprehensively analyze the tradeoff between our attack and the text quality using multiple automated studies.

Our core contributions are:

- We introduce **Adversarial Paraphrasing** — a universal and transferable attack that humanizes AI-generated text by guiding an off-the-shelf LLM, used as a paraphraser, to sample each token based on detection scores from an off-the-shelf trained AI text detector.
- We conduct extensive experiments demonstrating the attack’s effectiveness and transferability across 8 different types of state-of-the-art AI text detectors, highlighting its universality.
- We evaluate the trade-off between attack success and the resulting text quality using perplexity scores as well as automated ratings with GPT-4o to find that our attacks can significantly reduce detection rates when compared to prior evasion attacks with only a slight or no degradation in the text quality.

Our findings highlight important vulnerabilities of existing AI text detectors in the presence of adversaries.

2 Related Work

AI- Text Detection. Recent studies have demonstrated the effectiveness of training neural networks to classify AI-generated versus human-written text [10, 8, 26, 30, 34, 18]. For example, Solaiman *et al.* [30] employed a RoBERTa-based [20] classifier to distinguish between human-written and GPT-2-generated texts. Building on this idea, Li *et al.* [18] improves their classifier’s performance in the wild by gathering a diverse dataset, MAGE, for training their network. RADAR [8] adversarially trains their detector using a paraphraser in an iterative manner to make their detector robust to paraphrasing attacks. To reduce the computational overhead of training a network-based detector, various zero-shot detectors are proposed [22, 1, 6, 17]. These detectors use an off-the-shelf LLM to evaluate statistical properties of candidate texts, such as their entropy or log probability scores, to perform detection. For example, DetectGPT [22] observes that AI-generated text tends to lie in regions of negative curvature in the log-probability landscape, while Fast-DetectGPT [1] improves efficiency by introducing conditional probability curvature. Watermarking has also been explored as a means of identifying AI-generated text. Kirchenbauer *et al.* [14] introduced the KGW watermark scheme that partitions the token vocabulary into green and red lists, encouraging the model to sample more green tokens, thereby embedding detectable patterns in the generated text. While the KGW scheme uses previously generated tokens to seed a random number generator for token partitioning, Zhao *et al.* [36] proposed a variant of KGW, the Unigram watermark, which uses a fixed token partitioning for generation for provably demonstrating robustness. Kudithipudi *et al.* [16] developed a robust, distortion-free watermarking technique aimed at reducing distributional shifts caused by watermarking. More recently, Dathathri *et al.* [4] introduced SynthID, which utilizes tournament sampling to create a scalable watermarking solution for LLMs employing speculative decoding.

Attacks on AI Text Detectors. Sadasivan *et al.* [28] showed that AI text detectors can be fooled using paraphrasing attacks. While basic paraphrasing methods are sufficient to defeat early zero-shot and trained detectors, more robust detectors [8, 14, 36, 1] require recursive paraphrasing to be effectively bypassed. To this end, Krishna *et al.* [15] proposed DIPPER, a powerful T5-based [24] paraphrasing model that significantly enhances the effectiveness of such attacks. Sadasivan *et al.* [28] also introduced spoofing attacks, where human-written texts are manipulated to be misclassified as AI-generated, thereby increasing type-I errors. In addition, they analyzed the theoretical difficulty of AI text detection, highlighting a fundamental trade-off between type-I and type-II errors even for optimal detectors. The theoretical limits of AI-generated content watermarking have also been explored in other works [35, 27], which analyze the inherent challenges in maintaining robustness under attack. There also exist previous adversarial attack methods for breaking watermarking techniques. For example, Jovanovic *et al.* [12] proposed *Watermark Stealing*, a technique that learns the watermark signatures of a language model and uses this knowledge to both evade and spoof watermark-based detectors effectively. However, this framework specifically targets watermarking detectors and is not